

Rishab Dey

✉ rishabdey.cr7@gmail.com ☎ +49 176 47081657 📍 44227 Dortmund 🌐 www.linkedin.com/in/rishab-dey
📊 kaggle.com/cr7rishab 🐙 github.com/RishabDey 🔗 bit.ly/3QAcNMD (Notion)

SUMMARY

Data Scientist (M.Sc.) with 3+ years of experience as a Data Analyst, applying machine learning and statistical methods across research and industry settings. Skilled in predictive modeling and time-series analysis, and explainable AI. Experienced in building automated data pipelines and interactive dashboards to support data-driven decision-making. Developed multi-agent LLM pipelines for automated research and source-attributed report generation, with experience in Generative AI workflows and applications. Proficient in Python, SQL, and TensorFlow, with expertise in model evaluation, feature engineering, and interpretability.

WORK EXPERIENCE

- 02/2023 – 03/2026
Dortmund, Germany
- Data Analyst Student Assistant**, Fraunhofer-Institut für Materialfluss und Logistik (Sustainability and Circular Economy)
- Designed and deployed a secure multi-user analytics dashboard to replace manual Excel-based sustainability reporting, automating GHG and KPI calculations for 120 companies with over 1,500 multi-year data submissions.
 - Processed and analyzed international trade datasets on plastics and rubber flows to and from Germany, building interactive geospatial visualizations to identify material transfer patterns and cross-border logistics trends.
 - Designed and deployed a real-time environmental monitoring system integrating IoT sensor streams (Arduino) with a Python-based analytics and visualization layer, enabling live tracking of temperature, location, speed, and UV metrics to support Digital Product Passport (DPP) initiatives and enhance lifecycle transparency.
- Tools:** Python, Streamlit, Pandas, NumPy, Plotly, Folium, Seaborn, Matplotlib, Arduino MKR WIFI 1010, MKR Environmental shield
- 02/2019 – 03/2020
Mumbai, India
- Data Analyst Intern**, Breezeitech Services Pvt. Ltd.
- Analyzed inventory and product performance data (routers, CCTV systems) to evaluate cost efficiency, defect rates, and sales trends, delivering data-driven insights to support pricing and procurement decisions.
 - Performed exploratory and statistical analysis on monthly spending and product metrics, building visual dashboards and reports to identify performance patterns and optimization opportunities.
- Tools:** Python, NumPy, Pandas, Plotly, Matplotlib, Seaborn.

EDUCATION

- 2021 – Feb 2026
Potsdam, Germany
- Master of Science in Data Science**, [Universität Potsdam](https://www.uni-potsdam.de/) 🌐
- Master Thesis:** Explainable AI for Time Series Monitoring of Cocoyam Drying Using Hyperspectral Imaging
- Key coursework: Machine Learning, Statistical Learning for Big Data, Advanced Statistical Learning, Computer Vision, Optimization, Linear Models, Learning in Robotics, Inference
- 2014 – 2018
Mumbai, India
- Bachelor of Engineering (Electronics)**, University of Mumbai
- Key coursework: Artificial Intelligence, Embedded System Design, Digital Signal Processing, Signals & Systems, CMOS VLSI Design

SKILLS

Programming: Python, R, SQL (MySQL, PostgreSQL)

Libraries: NumPy, Pandas, Scikit-learn, TensorFlow, Statsmodels, Matplotlib, Seaborn, Plotly, SpaCy

Machine Learning & Statistics: Time Series Analysis, Statistical Modeling, Feature Engineering, Model Optimization, Model Evaluation, Explainable AI, Generative AI (LLMs, Prompt Engineering, RAG, LangChain, Transformers)

Visualization & BI: Streamlit, Power BI, Tableau, Looker Studio, Folium, Geospatial Analysis

Data Systems & MLOps: MLflow, Git, Data Pipeline Development, Data Transformation, Data Quality Management

Other: Web scraping (Selenium, BeautifulSoup, rvest), MATLAB, Adobe Photoshop, Figma, PLC Programming, Arduino

LANGUAGES

- English (C1)
- German (A2, actively improving towards B1/B2 for professional use)

PROJECTS

Source-Aware Multi-Agent Research Pipeline → [↗](#)

Developed a multi-agent system that generates structured research reports from a user query with explicit source attribution. Orchestrates agents for web search via Tavily API, content extraction (BeautifulSoup), report generation, and evaluation. Implemented a Streamlit interface with real-time pipeline tracking and downloadable outputs.

Tools: Python, LangChain, Mistral AI, Tavily API, BeautifulSoup, Streamlit

Explainable AI for Time Series Monitoring (Master's Thesis)

Developed a CNN-LSTM model to predict moisture content from hyperspectral near-infrared time-series data, achieving an R^2 of 0.87. Applied Grad-CAM and FusionGrad for model interpretability and feature attribution, enabling improved feature selection through the identification of key spectral bands and spatial regions.

Tools: Python, TensorFlow, NumPy, Pandas, Scikit-learn

WorldLytics : Interactive Data Insights → [↗](#)

Developing an evolving analytics portfolio featuring comparative insights across real-world datasets.

Visualization of Sports Data

Scraped data from *transfermarkt.de* on football players' performance, wages, and club spending across Europe's top 5 leagues. Visualized this data to have interesting insights, and authored a supporting report

Tools : R, rvest, Selenium, Python, Matplotlib, Seaborn, Plotly, NumPy, Pandas

Fashion MNIST Image Classification using Deep Learning & Dimensionality Reduction

Built a CNN-based image classifier on the Fashion-MNIST dataset and compared performance across multiple optimizers. Applied PCA, Kernel PCA, and t-SNE for dimensionality reduction and cluster visualization to analyze feature separability.

Tools used: Python, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, Kaggle

Geospatial Tracking the Spread of COVID-19

Visualized the spread of Covid-19 across India, Italy and South Korea. A state wise visualization was carried out for India, highlighting total cases, deaths, active cases and recoveries. Utilized the *folium* library for interactive geospatial mapping

Tools: Python, NumPy, Pandas, Matplotlib, Plotly, Seaborn, Folium

Comparative Analysis of Housing Prices

Used real estate data from a German real estate web portal, [Immoscout24.de](#) [↗](#) to compare rental prices across four German cities to see if the rental prices of these areas differ from each other. Analysis of variance (ANOVA) , t- test , and two sample test were carried out. Applied Bonferroni correction to address and rectify multiple testing issues. **Tools used:** R, dplyr, ggpubr, ggplot2

PAPERS AND PRESENTATIONS

Explainable AI for Time-Series Monitoring of Cocoyam Drying Using Hyperspectral Imaging

Manuscript in preparation for conference/journal submission (2026), Universität Potsdam.

Conference on Advances in Electronics Engineering (CAEE 2018),

Presented and published a paper on 'Waste Management System' - ISBN Number: 978-0-9994483-6-6

ACHIEVEMENTS

- Captain of Junior College Football Team,
- Taekwondo Black Belt 1st Dan , Sang Rok World, Recognized by World Taekwondo Federation
- First Price in Second Year Bachelors Project
- Vice-Captain of Degree College Football Team
- Rapporteur, International Press and Research Team (Model United Nations, held at the University)
- Third Price in Third Year Bachelors Project